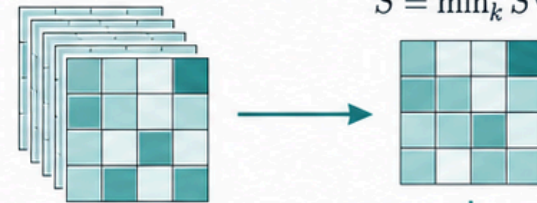


K safety budgets

$S = \min_k S^{(k)}$



shared capacities

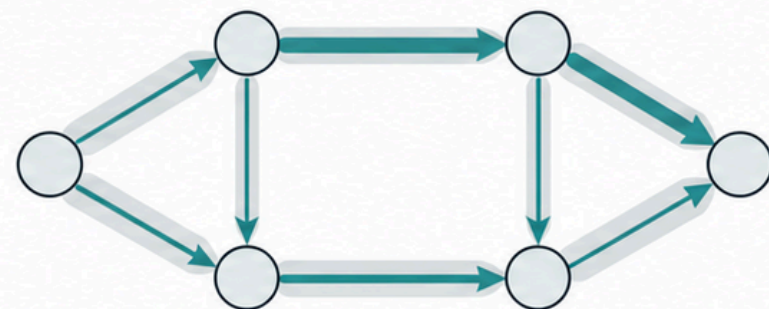
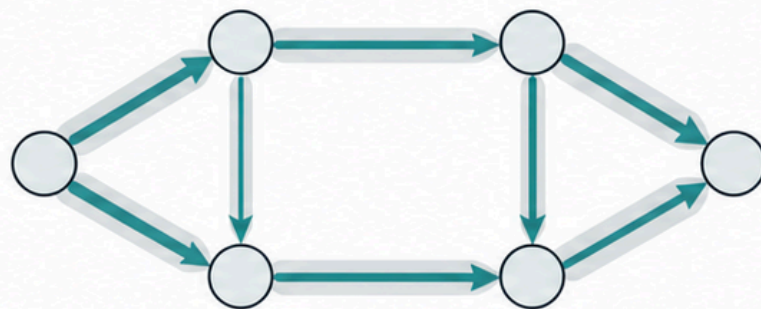
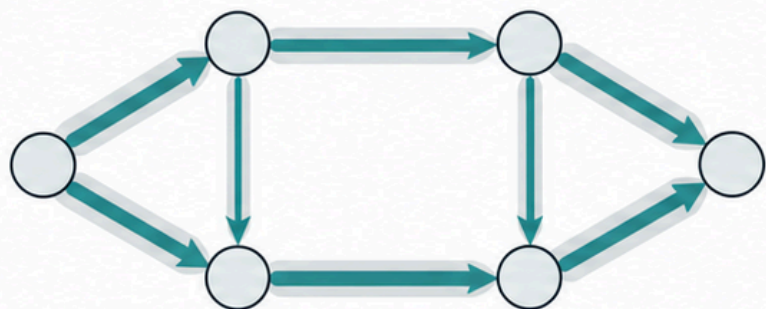
SafeTransport

One capacity-flow algorithm

$$F_\epsilon^* = \arg \min_{F \in \mathcal{F}_S} \langle D, F \rangle - \epsilon H(F)$$

$\mathcal{F}_S$: discounted flow balance, $0 \leq F \leq S$

gray: capacity
teal: flow



Penalty-like

larger ϵ · diffuse flow

vary ϵ within the same solver

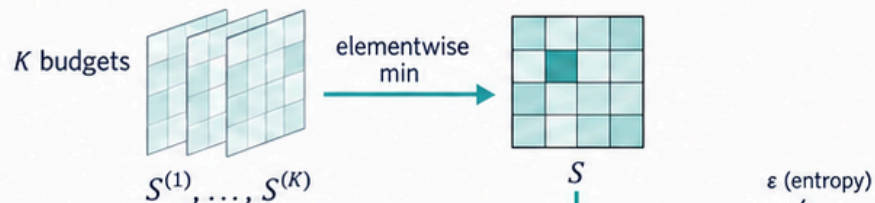
Constraint-like

smaller ϵ · concentrated flow

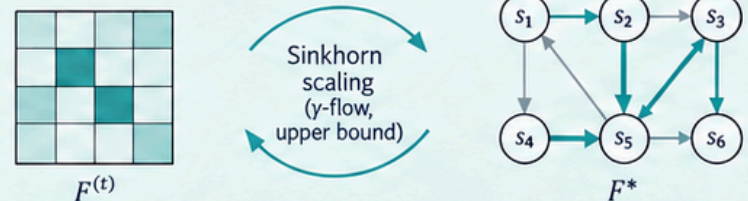
Shared representation across safe-RL regimes; capacities remain enforced.

samples, $\hat{F}_{\text{roll}}$

Compile budgets into capacities



Capacity-constrained γ -flow Sinkhorn



$$\min \langle D, F \rangle - \varepsilon H(F)$$

$$\text{discounted flow balance} \cdot 0 \leq F \leq S$$

Flow-weighted PPO

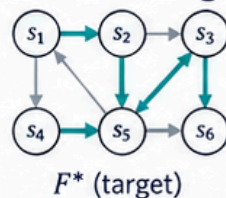
clipped flow-ratio weights



PPO → π_θ

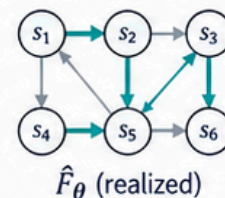
evaluate $\hat{F}_\theta$

Measure target-to-policy transfer



$$\Delta F = \|F^* - \hat{F}_\theta\|_1$$

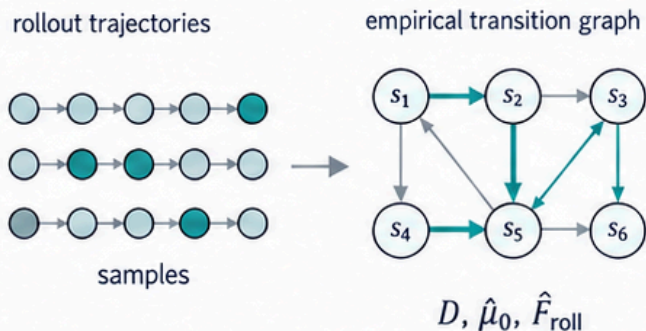
per-channel costs



persistent gap → recovery
SafeTransport-Safe
per-channel Lagrangian update

next rollout

Rollout → flow graph



graph statistics
 $D, \hat{\mu}_0$